

Requirements Analysis

A restaurant has decided to create a new information system with the following specifications. It is necessary to store information about all the restaurant's menu items, considering as relevant information the name of the menu item, price, category (e.g., appetizer, main course, dessert), and stock quantity available.

The restaurant manages customer reservations, each associated with a specific customer and a table. Customers can make multiple reservations, but a reservation is linked to only one customer and one table at a time. It is required to store the reservation date, time, and the number of guests for each reservation.

Customers can place orders during their visit, and each order can include multiple menu items. Each order is associated with a specific reservation, and it is necessary to store the total amount for the order and its status (e.g., completed, pending).

Additionally, customers can provide feedback on their dining experience, which includes a rating and a comment. Each customer can leave multiple feedback entries, but each feedback entry is linked to only one customer.

The restaurant also needs to manage tables, with relevant information including the table number and its capacity. Each table can have multiple reservations over time.

Furthermore, the restaurant has a Membership program that provides exclusive benefits to customers. Each membership is uniquely identified by a numeric Membership ID and is associated with a specific customer. Each customer can have only one membership, and each membership is linked to only one customer. The membership will include relevant information such as the membership start date, expiration date, and membership type. There are three types of memberships – “Gold”, “Silver” and “Bronze”.